Week 11

Course Level Objectives:

- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5) **2.** Construct a confidence interval for a population proportion using z* for 90%, 95%, or 99% confidence. (CLO 6) **3.** Interpret a confidence interval for a proportion in context. (CLO 6) **4.** Perform a one-proportion z test and state the conclusion in context. (CLO 8) **5.** Explain why the confidence-interval standard error uses $\hat{p}$ while the test uses p_0. (CLO 8)	e-Text: 6.1-CI Confidence Interval for a Proportion e-Text: 6.1-D Distribution of a Proportion e-Text: 6.1-HT Hypothesis Test for a Proportion Interactive Lecture: 6.1 Inference for a Proportion Slides: 6.1 Inference for a Proportion	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs Your Turn: Testing a Proportion (60 of 250 live on campus)	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM) Take-Home Quiz 7 (due Mon Nov 16, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus
6. Determine whether a study calls for one-proportion or difference-in-proportions inference. (CLO 5) **7.** Check the conditions for inference about a difference in proportions. (CLO 5) **8.** Construct a confidence interval for $p_1 - p_2$. (CLO 6) **9.** Interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CLO 6) **10.** Compute the pooled proportion. (CLO 8) **11.** Perform a two-proportion z test from summary data or a two-way table. (CLO 8)	e-Text: 6.3-CI Confidence Interval for a Difference in Proportions e-Text: 6.3-D Distribution of a Difference in Proportions e-Text: 6.3-HT Hypothesis Test for a Difference in Proportions Interactive Lecture: 6.3 Inference for a Difference in Proportions Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	Participation: Difference in Proportions – two-proportion randomization (18/30 vs. 12/30); compare simulation with the formula Your Turn: Treatment vs Placebo (two-proportion z test from a two-way table)	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 8, 11:59 PM) Take-Home Quiz 7 (due Mon Nov 16, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus

Also this week: Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)